Karthik Gnana Ezhil S

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.48
Paavai Vidhyashram	Class XII (CBSE)	2023	91.80%
Paavai Vidhyashram	Class X (CBSE)	2021	92.16%

SCHOLASTIC ACHIEVEMENTS

Qualified **JEE Mains** and **JEE Advanced 2023**, among India's most competitive entrance exams (1.2 million+ and 1.8 lakh+ candidates respectively)

Qualified for **NTSE Stage 2** (Top ~1% nationwide · 2020)

SKILLS

Programming Languages: C, C++, Java, JavaScript, Python, dart

Web Development: React, Node.js, Express.js, FastAPI, GraphQL, SQL (PostgreSQL), MongoDB

Tools & Platforms: Azure, GCP, AWS, Docker

LLM & Agentic AI: RAG, LangGraph, Claude Agent SDK, MCP, Fine-tuning, AWS SageMaker

Machine Learning: PyTorch, Tensorflow, Transformers, SBERT, Deep Learning

App Development: Flutter and Dart, Android Native

Problem solving: Data Structures & Algorithms **(400+ LeetCode & 100+ GfG problems)**

Soft Skills: Leadership, Teamwork and Communication

RELEVANT COURSES

COURSES:

- Data Structures and Algorithms
- Introduction to Programming
- Machine Learning for Physical Sciences

WORK EXPERIENCE

Applied Research Intern | Thomson Reuters

Publications: *Teaching a Small Model to Check a Big Model's Work -- (Medium)*

- Architected an **agentic** tax onboarding system using **Agent Skills, MCPs,** and **human-in-the-loop workflows**, cutting onboarding from **weeks** to **~1 hour**.

- Developed a **multi-turn simulation** and **LLM-as-a-judge** evaluation framework to test the agent and iteratively improve its production accuracy.
- **Fine-tuned 0.6B & 4B LLMs** for retrieval reranking and response self-evaluation, matching frontier proprietary LLM performance on select metrics.
- Automated large-scale **benchmarking** of **4B-27B open-source LLMs** via Claude skills orchestrating **vLLM inference** and **GPU parallelism** across **multi-GPU** infra.

PROJECTS

InsIIT - Q&A Forum for JEE Aspirants

Tech Stack: *React js, Express Js, MongoDB, GraphQL*

Link: https://insiit.in/

- Built and Deployed Q&A forum on **Azure VM**, handling server setup, configuring networking, security rules, and uptime monitoring.
- Implemented **Google OAuth2** and enforced **role-based access** for aspirants and campus experts.
- Achieved **400+ users** who registered and posted over **50 questions** in the **first week**.

E-Commerce Backend using Java and Spring Boot

Tech Stack: *Spring Boot Web, PostgreSQL, Redis*

GitHub: https://github.com/SKGEzhil/E-Commerce-Backend

- Developed a **Spring Boot RESTful API** backend, containerized it with **Docker** and deployed to **Azure Container App**
- Implemented **OAuth2 + JWT-based** API security and used **Redis caching** to reduce latency by **30%**.
- Implemented **Stripe payment** integration with idempotent webhooks for reliable, retry-safe transactions.

College Email Categorization Model:

Tech Stack: *Python, Pandas, Transformer, Tensorflow, Docker, FastAPI*

GitHub: https://github.com/SKGEzhil/email-classifier

- Built a **Transformer** based classification model using **SBERT** to auto-label college emails with **94% accuracy** on **5000 samples.**
- **Deployed** model server on **Azure Container Apps** and Gmail listener server on **AWS AppRunner** for **automated** labelling of incoming emails.
- Developed GitHub Actions **CI/CD pipelines** to build, test, and **auto-deploy** both Docker services to Azure and AWS on each commit.

JAXRLAgent -- High Performance Reinforcement Learning Library

Tech Stack: *JAX, Flax NNX, Optax, Orbax, WandB, Python*

- Built a RL library from scratch in **JAX** achieving **up to 10× faster training** than standard PyTorch baselines across multiple tasks
- Implemented **6 RL algorithms** and **5 configurable neural network architectures**, unified under a single modular

framework

- Built a **config driven benchmark suite** for testing of any environment across multiple agent-network configurations

AI Expense -- Smart Expense Tracker

Tech Stack: *Python, Flutter, PostgreSQL, Vector DB, Finetuning*

GitHub: https://github.com/SKGEzhil/ai_expense

- Built an **AI-powered** expense tracker that extracts transaction details automatically from payment screenshots, eliminating manual data entry
- Integrated **pgvector** into PostgreSQL, enabling **semantic search** across transactions using **vector embeddings**
- **Fine-tuned** a **270M** parameter SLM for translating **natural language into SQL** by baking schema into training, cutting inference cost while improving searchability

POSITIONS OF RESPONSIBILITY

Core App Developer - **Lambda**

- Collaborated on the IITH Dashboard App using **Flutter & Dart**, contributing **UI components** and integrating **RESTful APIs**

Web Development Head - Alumni Cell

- Lead a 4-member dev team to define the roadmap and technical architecture for the new ACR website

EXTRACURRICULAR

Head - BTL (Film Making Club), IIT Hyderabad

- Lead a 13-member team through concept development, scheduling, shooting, and post-production.